

concentration during exercise of increasing intensity. Generally, individuals with higher lactate thresholds, expressed as a percentage of their $\dot{V}O_{2\max}$, are capable of better endurance performances. Lactate threshold is a strong determinant of an athlete's optimal pace in endurance events such as distance running and cycling.

TABLE 5.2 Average Values for Energy Expenditure During Various Physical Activities

Activity	Men (kcal/min)	Women (kcal/min)	Relative to body mass (kcal · kg ⁻¹ · min ⁻¹)
Basketball	8.6	6.8	0.123
Cycling			
11.3 km/h (7.0 mph)	5.0	3.9	0.071
16.1 km/h (10.0 mph)	7.5	5.9	0.107
Handball	11.0	8.6	0.157
Running			
12.1 km/h (7.5 mph)	14.0	11.0	0.200
16.1 km/h (10.0 mph)	18.2	14.3	0.260
Sitting	1.7	1.3	0.024
Sleeping	1.2	0.9	0.017
Standing	1.8	1.4	0.026
Swimming (crawl), 4.8 km/h (3.0 mph)	20.0	15.7	0.285
Tennis	7.1	5.5	0.101
Walking, 5.6 km/h (3.5 mph)	5.0	3.9	0.071
Weightlifting	8.2	6.4	0.117
Wrestling	13.1	10.3	0.187

Note. The values presented are for a 70 kg (154 lb) man and a 55 kg (121 lb) woman. These values will vary depending on individual differences.

Fatigue and Its Causes

The term **fatigue** means different things to different people. Sensations that exercising individuals describe as fatigue are markedly different for a 400 m (437 yd) runner (an event lasting 45 to 60 s) than for a marathoner nearing the end of a 42.2 km (26.2 mi) endurance event. Therefore, it is not surprising that the causes of fatigue are different in those two scenarios as well. In exercise physiology, we typically describe *fatigue* as decrements in muscular performance with continued effort accompanied by general sensations of tiredness. An alternative definition used in research studies to quantify fatigue is the inability to maintain the required power output to continue muscular work at a given intensity. The fact that fatigue is reversible by rest distinguishes it from muscle weakness or damage (discussed later in the chapter).

RESEARCH PERSPECTIVE 5.1

Energy Expenditure of Walking

Understanding how much metabolic energy is expended during walking has many applications, from clinical rehabilitation programs to fitness and activity tracking and even military maneuvers. The metabolic energy required for walking can be accurately determined by directly measuring oxygen consumption during activity. However, this measurement technique is impractical outside the laboratory. Published equations that predict this energy requirement are frequently used. The two most established and commonly used equations are those developed by the American College of Sports Medicine (ACSM)¹ and by a group of research scientists at the U.S. Army Research Institute of Environmental Medicine (USARIEM).²¹ Both of these equations are specific to body mass and divide the person's metabolic rate into resting and nonresting components, where the nonresting component is speed dependent. Although these equations are considered the gold standards for predicting energy expenditure during walking, they were developed based on studies that included only young, healthy, male subjects of a relatively similar body size.

A recent study conducted at Southern Methodist University in Dallas, Texas, derived a new mathematical model to predict the metabolic energy requirements of walking.¹⁵ Data from 10 previous studies were compiled to create a data set of over 400 subjects of both sexes who varied in age, height, body weight, and fitness level. Researchers then developed mathematical models to identify the variables required for accurate predictions of metabolic energy requirements across this heterogeneous subject pool and found that the most accurate predictions accounted for the walker's height (ht), a variable that is absent in the commonly used equations. Like previous equations, that accuracy of the prediction was increased when the walking metabolism was quantified as two separate components, the minimum walking component (different from the resting component) and the velocity-dependent component. The study resulted in a new model for predicting metabolic energy expenditure that predicted over 90% of the actual metabolic cost across all walking speeds:

$$\dot{V}O_2 \text{ total} = (\dot{V}O_2 \text{ rest} + 3.85) + (5.97 \cdot v^2/\text{ht})$$

where walking velocity (v) is measured in m/s, ht in m, and $\dot{V}O_2$ in ml $O_2 \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$. In this equation, $(\dot{V}O_2 \text{ rest} + 3.85)$ is the minimum walking energy expenditure, $(5.97 \cdot v^2/\text{ht})$ is the velocity- and height-dependent energy expenditure, and $[3.85 + (5.97 \cdot v^2/\text{ht})]$ quantifies the total walking component.

This model can now be used to inform exercise prescriptions for numerous health and fitness outcomes in wide-ranging populations.

Ask most exercisers what causes fatigue during exercise, and the most common two-word answer is “lactic acid.” Not only is this common misconception an oversimplification but also there is mounting evidence that lactic acid has beneficial effects on exercise performance (see [chapter 2](#)). Fatigue is an extremely complex phenomenon, and its causes can range from the molecular level to the entire body. Most efforts to describe the underlying causes and sites of fatigue have focused on

- a decreased rate of energy delivery (ATP-PCr, anaerobic glycolysis, and oxidative metabolism);
- accumulation of metabolic by-products, such as lactate and H^+ ;
- failure of the muscle fiber’s contractile mechanism; and
- alterations in neural control of muscle contraction.

The first three causes occur within the muscle itself; along with alterations in motor nerve control of muscle function, these are often referred to as *peripheral* fatigue. In addition to alterations at the motor unit level, changes in the brain or central nervous system may also cause what has become known as *central* fatigue.

However, none of these alone can explain all aspects or all types of fatigue, and several causes may act synergistically to bring about fatigue. Mechanisms of fatigue depend on the type and intensity of the exercise, the fiber type of the involved muscles, the subject’s training status, and even his or her diet. Many questions about fatigue remain unanswered, including the cellular sites of fatigue within the muscle fibers themselves. It is important to remember that while fatigue arises at least in part from failure of cross-bridge cycling within the muscle cells, this machinery depends on the nervous, cardiovascular, and energy systems to support it.¹⁴ Fatigue is rarely caused by a single factor but typically by multiple factors acting synergistically at multiple sites. Some potential sites of fatigue are discussed next.

Energy Systems and Fatigue

The energy systems are an obvious area to explore when one is considering possible causes of fatigue. When we feel fatigued, we

often express this by saying, “I have no energy.” But this use of the term *energy* is far removed from its physiological meaning. What role does energy availability play in fatigue during exercise, in the true physiological sense of providing ATP from substrates?

PCr Depletion

Recall that PCr is used for short-term high-intensity effort, to rebuild ATP as it is used and thus to maintain ATP stores within the muscle. Biopsy studies of human thigh muscles have shown that during repeated maximal contractions, fatigue coincides with PCr depletion. Although ATP is directly responsible for the energy used during such activities, it is depleted less rapidly than PCr during muscular effort because ATP is being produced by other systems (see [figure 2.6](#)). But as PCr is depleted, the ability to quickly replace the spent ATP is hindered. Use of ATP continues, but the ATP-PCr system is less able to replace it. Thus, ATP concentration also decreases. At exhaustion, both ATP and PCr may be depleted.

To delay fatigue, an athlete must control the rate of effort through proper pacing to ensure that PCr and ATP are not prematurely exhausted. This holds true even in endurance-type events. If the beginning pace is too rapid, available ATP and PCr concentrations will quickly decrease, leading to early fatigue and an inability to maintain the pace in the event’s later stages. Training and experience allow the athlete to judge the optimal pace that permits the most efficient use of ATP and PCr for the entire event.

Glycogen Depletion

Muscle ATP concentrations are also maintained by the breakdown of muscle glycogen. In events lasting longer than a few seconds, muscle glycogen becomes the primary energy source for ATP synthesis. Unfortunately, glycogen reserves are limited and are depleted quickly. Since the muscle biopsy technique was first established, studies have shown a correlation between muscle glycogen depletion and fatigue during prolonged exercise.

Muscle glycogen is used more rapidly during the first few minutes of exercise than in the later stages, as seen in [figure 5.8](#).⁶ The illustration shows the change in muscle glycogen content in the subject’s gastrocnemius (calf) muscle during the test. Although the

subject ran the test at a steady pace, the rate of muscle glycogen metabolized from the gastrocnemius was greatest during the first 75 min.

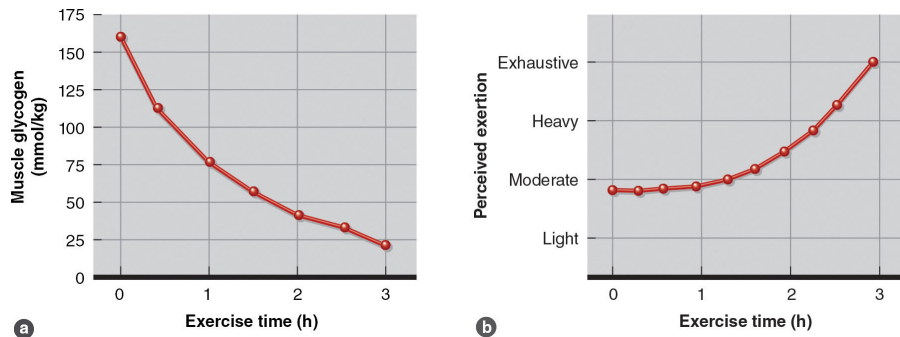


FIGURE 5.8 (a) The decline in gastrocnemius (calf) muscle glycogen during 3 h of treadmill running at 70% of $\dot{V}O_{2\max}$, and (b) the subject's subjective rating of the effort. Note that the effort was rated as moderate for nearly 1.5 h of the run, although glycogen was decreasing steadily. Not until the muscle glycogen became quite low (less than 50 mmol/kg) did the rating of effort increase.

Adapted by permission from D.L. Costill, *Inside Running: Basics of Sports Physiology* (Indianapolis: Benchmark Press, 1986). Copyright 1986 Cooper Publishing Group, Carmel, IN.



The subject also reported his perceived exertion (how difficult his effort seemed to be) at various times during the test. He felt only moderately stressed early in the run, when his glycogen stores were still high, even though he was using glycogen at a high rate. He did not perceive severe fatigue until his muscle glycogen levels were nearly depleted. Thus, the sensation of fatigue in long-term exercise coincides with a decreased concentration of muscle glycogen but not with its rate of depletion. Marathon runners commonly refer to the sudden onset of fatigue that they experience at 29 to 35 km (18-22 mi) as “hitting the wall.” At least part of this sensation can be attributed to muscle glycogen depletion.

Glycogen Depletion in Different Fiber Types

Muscle fibers are recruited and deplete their energy reserves in selected patterns. The individual fibers most frequently recruited

during exercise may become depleted of glycogen. This reduces the number of fibers capable of producing the muscular force needed for exercise.

This glycogen depletion is illustrated in [figure 5.9](#), which shows a micrograph of muscle fibers taken from a runner after a 30 km (18.6 mi) run. [Figure 5.9a](#) has been stained to differentiate type I and type II fibers. One of the type II fibers is circled. [Figure 5.9b](#) shows a second sample from the same muscle, stained to show glycogen. The redder (darker) the stain, the more glycogen is present. Before the run, all fibers were full of glycogen and appeared red (not depicted). In [figure 5.9b](#) (after the run), the lighter type I fibers are almost completely depleted of glycogen. This suggests that type I fibers are used more heavily during endurance exercise that requires only moderate force development, such as the 30 km run.

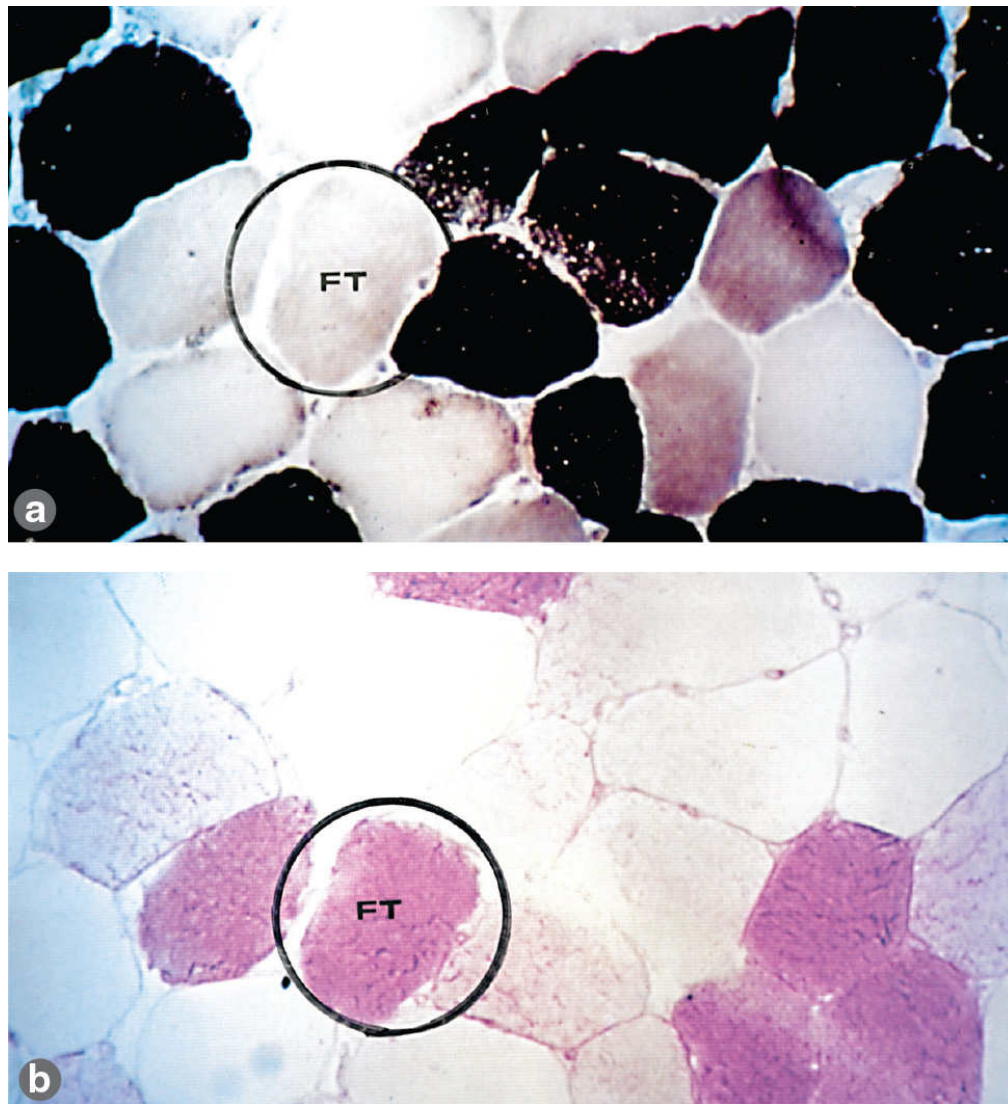


FIGURE 5.9 (a) Histochemical staining for fiber type after a 30 km run; a type II (fast-twitch) fiber is circled. (b) Histochemical staining for muscle glycogen after the run. Note that a number of type II fibers still have glycogen, as noted by their darker stain, whereas most of the type I (slow-twitch) fibers are depleted of glycogen.

The pattern of glycogen depletion from type I and type II fibers depends on the exercise intensity. Recall that type I fibers are the first fibers to be recruited during light exercise. As muscle tension requirements increase, type IIa fibers are added to the workforce. In exercise approaching maximal intensities, the type IIx fibers are added to the pool of recruited fibers.

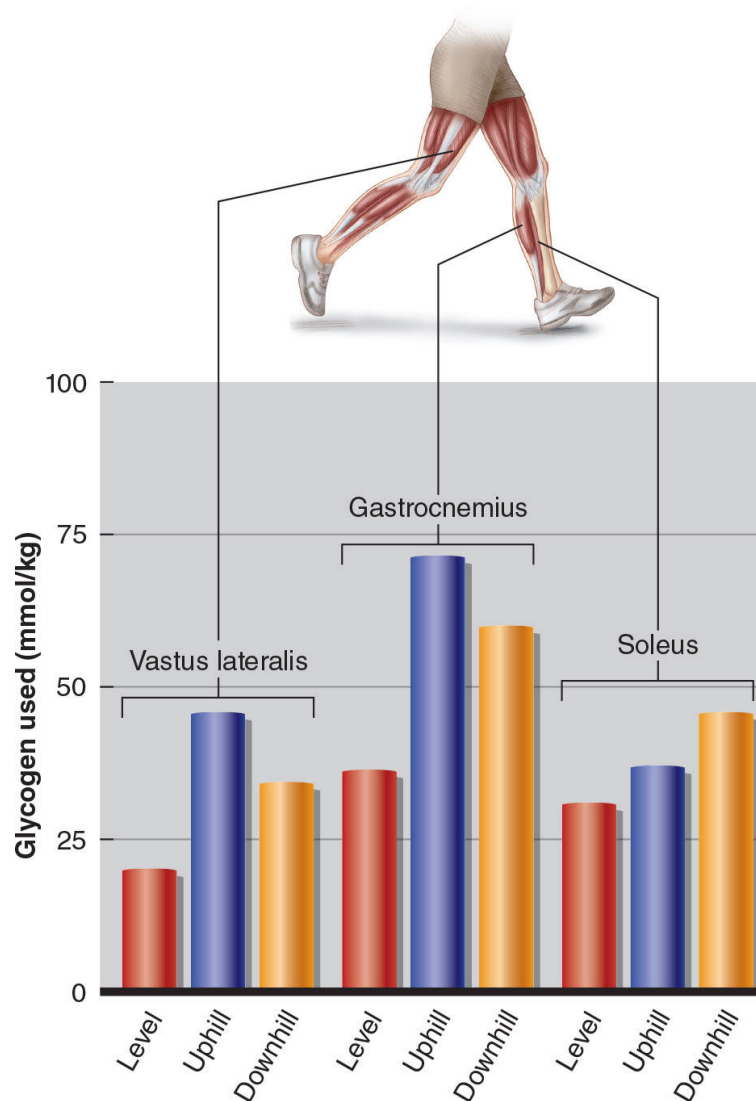


FIGURE 5.10 Muscle glycogen use in the vastus lateralis, gastrocnemius, and soleus muscles during 2 h of level, uphill, and downhill running on a treadmill at 70% of $\dot{V}O_{2\max}$. Note that the greatest glycogen use is in the gastrocnemius during uphill and downhill running.



Depletion in Different Muscle Groups In addition to selectively depleting glycogen from type I or type II fibers, exercise may place unusually heavy demands on select muscle groups. In one study, subjects ran on a treadmill positioned for uphill, downhill, and level running for 2 h

at 70% of $\dot{V}O_{2\max}$. [Figure 5.10](#) compares the resultant glycogen depletion in three muscles of the lower extremity: the vastus lateralis (knee extensor), the gastrocnemius (ankle extensor), and the soleus (another ankle extensor).

The results show that whether one runs uphill, downhill, or on a level surface, the gastrocnemius uses more glycogen than does the vastus lateralis or the soleus. This suggests that the ankle extensor muscles are more likely to become depleted during distance running than are the thigh muscles, isolating the site of fatigue to the lower leg muscles.

Glycogen Depletion and Blood Glucose Muscle glycogen alone cannot provide enough carbohydrate for exercise lasting several hours. Glucose delivered by the blood to the muscles contributes a lot of energy during endurance exercise. The liver breaks down its stored glycogen to provide a constant supply of blood glucose. In the early stages of exercise, energy production requires relatively little blood glucose, but in the later stages of an endurance event, blood glucose may make a large contribution. To keep pace with the muscles' glucose uptake, the liver must break down increasingly more glycogen as exercise duration increases.

Liver glycogen stores are limited, and the liver cannot produce glucose rapidly from other substrates. Consequently, blood glucose concentration can decrease when muscle uptake exceeds the liver's glucose output. Unable to obtain sufficient glucose from the blood, the muscles must rely more heavily on their glycogen reserves, accelerating muscle glycogen depletion and leading to earlier exhaustion.

Not surprisingly, endurance performances improve when the muscle glycogen supply is elevated before the start of activity. On the other hand, most studies have shown no effect of carbohydrate ingestion on net muscle glycogen utilization during prolonged, strenuous exercise. The importance of muscle glycogen storage for endurance performance is discussed in [chapter 15](#). For now, note that glycogen depletion and hypoglycemia (low blood sugar) limit performance in activities lasting longer than 60 min.

Mechanisms of Fatigue with Glycogen Depletion It does not appear likely that glycogen depletion *directly* causes fatigue during endurance exercise performance, but it may play an *indirect* role. We cannot explain precisely why muscle function is impaired when muscle glycogen is low, but this is usually explained by a compromised rate of ATP production. Glycogen is more than simply a form for carbohydrate storage; it also acts as a regulator of several cellular functions. To aid in that role, glycogen is not distributed homogeneously throughout the muscle fiber but localized in distinct pools. Evidence suggests that depletion of glycogen granules localized within the myofibrils interferes with excitation–contraction coupling and Ca^{2+} release from the sarcoplasmic reticulum.²⁰

Metabolic By-Products and Fatigue

Various by-products of metabolism have been implicated as factors causing, or contributing to, fatigue. The metabolic by-products that have received the most attention in discussions of fatigue are inorganic phosphate, heat, lactate, and hydrogen ions.

Inorganic Phosphate

Inorganic phosphate increases during intense short-term exercise as PCr and ATP are being broken down. It now appears that P_i , which accumulates during intense short-term exercise from the breakdown of ATP, may be the largest contributor to fatigue in this type of exercise.²⁶ Excess P_i directly impairs contractile function of the myofibrils and can reduce Ca^{2+} release from the sarcoplasmic reticulum. Increases in both P_i and ADP also inhibit ATP breakdown through negative feedback.

Heat and Muscle Temperature

Recall that energy expenditure results in a relatively large heat production, some of which is retained in the body, causing core temperature to rise. Exercise in the heat can increase the rate of carbohydrate utilization and hasten glycogen depletion, effects that may be stimulated by the increased secretion of epinephrine. It is hypothesized that high muscle temperatures impair both skeletal muscle function and muscle metabolism.

The ability to continue moderate- to high-intensity cycle performance is affected by ambient temperature. Galloway and Maughan¹² studied performance time to exhaustion of male cyclists at four different air temperatures: 4 °C (38 °F), 11 °C (51 °F), 21 °C (70 °F), and 31 °C (87 °F). Results of that study are shown in [figure 5.11](#). Time to exhaustion was longest when subjects exercised at an air temperature of 11 °C and was shorter at colder and warmer temperatures. Fatigue set in earliest at 31 °C. Similarly, at a given warm air temperature, increasing relative humidity caused early fatigue.¹⁶ Precooling of muscles similarly prolongs exercise, while preheating causes earlier fatigue. Heat acclimation, discussed in [chapter 12](#), spares glycogen and reduces lactate accumulation.

Lactic Acid

Recall that lactic acid is a by-product of anaerobic glycolysis. Although most lay people believe that lactic acid is responsible for fatigue in all types of exercise, lactic acid undergoes constant turnover and, as described in [chapter 2](#), is recycled to provide energy. Lactic acid produced within the cytoplasm of a muscle fiber can be taken up by mitochondria within that same muscle fiber and oxidized for ATP formation. Lactic acid can also be shuttled to other sites where it can be oxidized. In fact, lactic acid only accumulates within a muscle fiber during relatively brief, highly intense muscular effort. Marathon runners often have near-baseline lactic acid concentrations at the end of the race, despite notable fatigue. As noted in the previous section, their fatigue is likely caused by inadequate energy supply, not excess lactic acid.

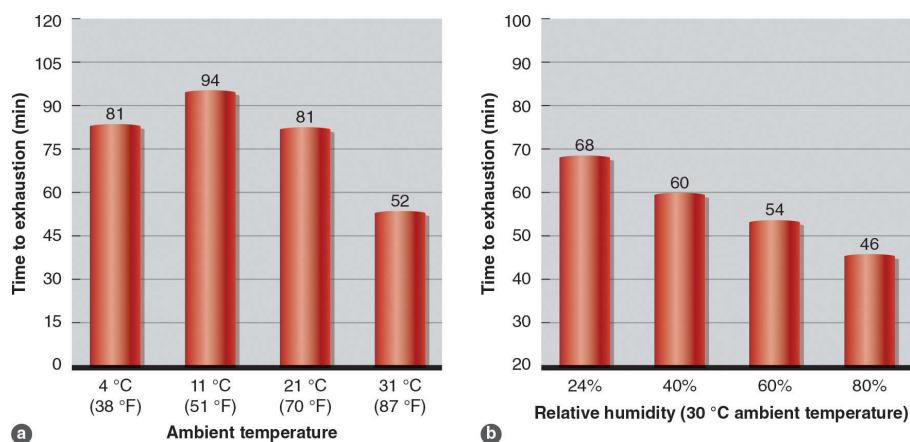


FIGURE 5.11 Time to exhaustion for a group of men performing cycle exercise at about 70% $\dot{V}O_{2\max}$. (a) The subjects were able to perform longer (delay fatigue longer) in a cool environment of 11 °C. Exercising in colder or warmer conditions hastened fatigue. (b) At an ambient temperature of 30 °C, increased relative humidity decreased time to exhaustion.

(a) Adapted by permission from S.D.R. Galloway and R.J. Maughan, "Effects of Ambient Temperature on the Capacity to Perform Prolonged Cycle Exercise in Man," *Medicine and Science in Sports and Exercise* 29 (1997): 1240-1249. (b) Reprinted by permission from R.J. Maughan et al., "Influence of Relative Humidity on Prolonged Exercise Capacity in a Warm Environment," *European Journal of Applied Physiology* 112 (2012): 2313-2321.

Short sprints in running, cycling, and swimming can all lead to large accumulations of lactic acid. While the presence of lactic acid in itself cannot be blamed for the feeling of fatigue, if it is not cleared, the lactic acid dissociates, converting to lactate and causing an accumulation of hydrogen ions.

Hydrogen Ions

While the lactate ion does not appear to have any major negative effects on the ability to generate force, H^+ accumulation causes muscle acidosis (decreased pH).

Activities of short duration and high intensity, such as sprint running and sprint swimming, depend heavily on anaerobic glycolysis and produce large amounts of lactate and H^+ within the muscles. Fortunately, the cells and body fluids possess buffers, such as bicarbonate (HCO_3^-), that minimize the disrupting influence of the H^+ . Without these buffers, H^+ would lower the pH to about 1.5, killing the cells. Because of the body's buffering capacity, the H^+ concentration remains low even during the most severe exercise, allowing muscle pH to decrease from a resting value of 7.1 to no lower than 6.4 at exhaustion after high-intensity activity.

However, pH changes of this magnitude can adversely affect energy production and muscle contraction. An intracellular pH below 6.9 inhibits the action of phosphofructokinase, an important glycolytic enzyme, slowing the rate of glycolysis and ATP production. At a pH of 6.4, the influence of H^+ stops any further glycogen breakdown, causing a rapid decrease in ATP and ultimately exhaustion. In addition, H^+ may lower the amount of calcium released from the sarcoplasmic reticulum, interfering with the coupling of the actin-myosin cross-bridges and decreasing the muscle's contractile force. However, the impact on muscle force production is small. A larger

impact comes from H^+ acting to decrease the myofilaments' sensitivity to calcium, causing a loss of contractile force and velocity.⁸ Because of those effects, low muscle pH may be a primary cause of fatigue during maximal, all-out exercise lasting 20 to 30 s.

As seen in [figure 5.12](#), reestablishing the preexercise muscle pH after an exhaustive sprint bout requires about 30 to 35 min of recovery. Even when normal pH is restored, blood and muscle lactate levels can remain quite elevated. However, experience has shown that an athlete can continue to exercise at relatively high intensities even with a muscle pH below 7.0 and a blood lactate level above 6 or 7 mmol/L, four to five times the resting value.

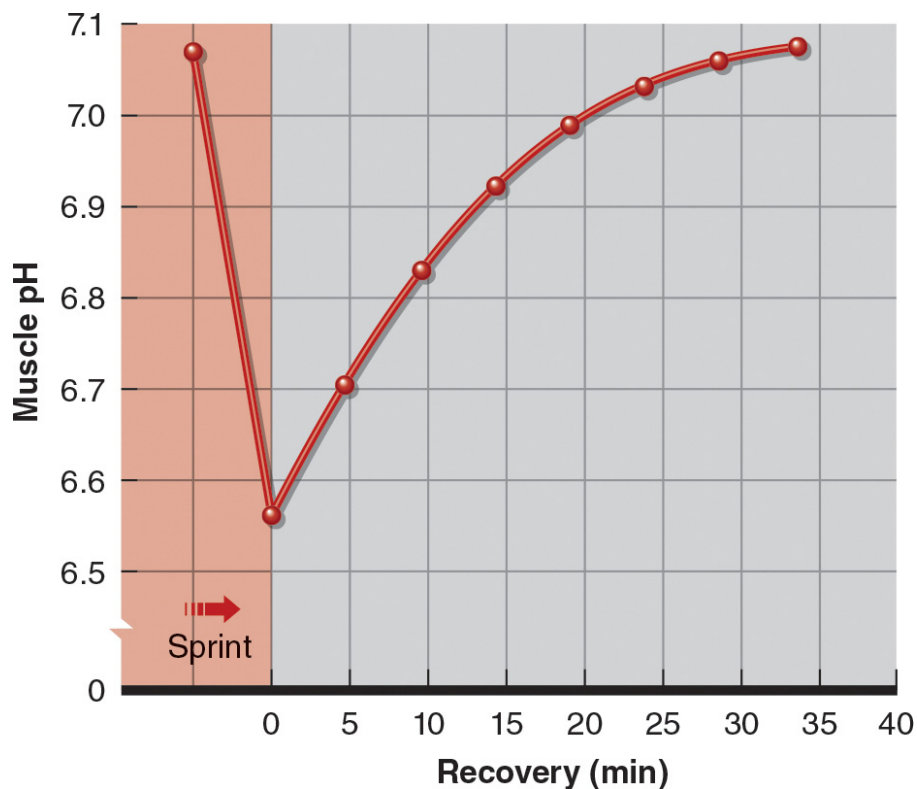


FIGURE 5.12 Changes in muscle pH during sprint exercise and recovery. Note the drastic decrease in muscle pH during the sprint and the gradual recovery to normal after the effort. Note that it took more than 30 min for pH to return to its preexercise level.



Neuromuscular Fatigue

Thus far we have considered only factors within the muscle that might be responsible for fatigue. Evidence also suggests that under some circumstances, fatigue may result from an inability to activate the muscle fibers, a function of the nervous system. As noted in [chapter 3](#), the nerve impulse is transmitted across the neuromuscular junction to activate the fiber's membrane, and it causes the fiber's sarcoplasmic reticulum to release calcium. The calcium, in turn, binds with troponin to initiate muscle contraction, a process collectively called excitation–contraction coupling. Several possible neural mechanisms could disrupt this process and possibly contribute to fatigue, and two of those—one peripheral and one central—are discussed next.

Neural Transmission

Fatigue may occur at the neuromuscular junction, preventing nerve impulse transmission to the muscle fiber membrane. Studies in the early 1900s clearly established such a failure of nerve impulse transmission in fatigued muscle. This failure may involve one or more of the following processes:

- The release or synthesis of acetylcholine (ACh), the neurotransmitter that relays the nerve impulse from the motor nerve to the muscle membrane, might be reduced.
- Cholinesterase, the enzyme that breaks down ACh once it has relayed the impulse, might become hyperactive, preventing sufficient concentration of ACh to initiate an action potential.
- Cholinesterase activity might become hypoactive (inhibited), allowing ACh to accumulate excessively, inhibiting relaxation.
- The muscle fiber membrane might develop a higher threshold for stimulation by motor neurons.
- Some substance might compete with ACh for the receptors on the muscle membrane without activating the membrane.
- Potassium might leave the intracellular space of the contracting muscle, decreasing the membrane potential to half of its resting value.

Although most of these causes for a neuromuscular block have been associated with neuromuscular diseases (such as myasthenia gravis), they may also cause some forms of neuromuscular fatigue. Some evidence suggests that fatigue also may be attributable to calcium retention within the sarcoplasmic reticulum, which would decrease the calcium available for muscle contraction. In fact, depletion of PCr and lactate buildup might simply increase the rate of calcium accumulation within the sarcoplasmic reticulum. However, these theories of fatigue remain speculative.

Central Nervous System

The discussion to this point suggests that fatigue is due to *peripheral* changes that limit or completely stop further effective muscular actions. The recruitment of muscle depends, in part, on conscious or subconscious control by the brain. An alternate theory to peripheral fatigue, termed the **central governor theory**, proposes that processes occur in the brain that regulate power output by the muscles to maintain homeostasis and prevent unsafe levels of exertion that may damage tissues or cause catastrophic events. The central governor limits exercise by decreasing the recruitment of muscle fibers, which in turn causes fatigue. While this theory has been hotly debated in recent years, the concept of a central “governor” was first proposed by A.V. Hill (see introductory chapter) in 1924.

In a 2012 study, researchers in Switzerland sought to separate out the central and peripheral contributors to muscle fatigue during low-intensity isometric contraction of the knee extensor muscles using an innovative protocol.¹⁸ Subjects performed a sustained isometric muscle contraction at 20% of the maximal voluntary contraction (MVC) until they experienced fatigue, defined as the point at which they could no longer maintain the 20% MVC force output. Then, the muscle was immediately electrically stimulated to maintain the same force output for 1 min, followed by an immediate voluntary effort to maintain that same force once again. In essence, fatigue was induced, and then the external electrical stimulation took over for the brain and the motor neuron to continue to produce force. If the initial fatigue was caused by problems with excitation–contraction coupling

(peripheral factors), then electrically stimulated muscle would still be fatigued and not able to produce force. Conversely, if fatigue was due to problems with the motor neuron or central neural factors, electrical stimulation would cause the muscle to generate force once again. The researchers also measured the maximal amount of force that could be voluntarily generated before and after fatigue was induced.

RESEARCH PERSPECTIVE 5.2

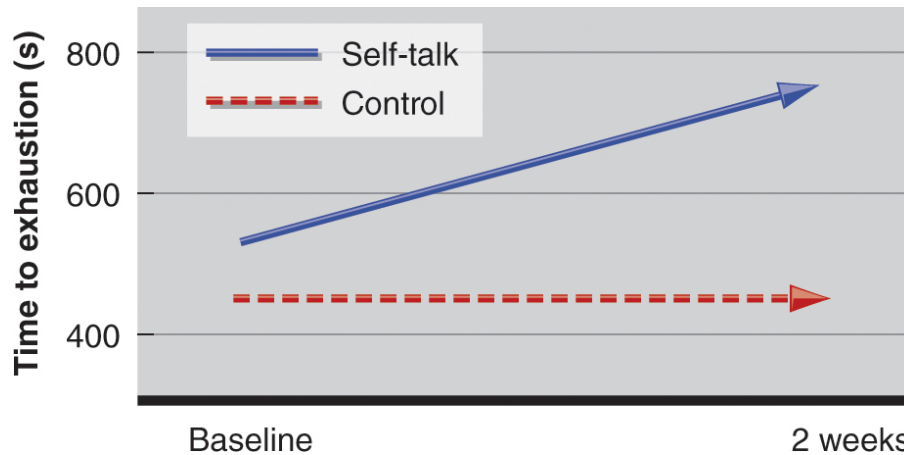
Can You Talk Yourself Out of Fatiguing?

As endurance sports become increasingly popular, the number of people participating in competitive endurance events is growing. Successful performance in endurance events requires the ability to sustain aerobic exercise over an extended period of time (that is, to delay the onset of fatigue as long as possible), and performance-enhancing strategies to increase the intensity and duration of the activity are important for athletes in these sports. Because fatigue defines the upper limit of endurance, research efforts have targeted understanding the physiological and psychological causes of fatigue and how to delay their effects.

During long-duration endurance events such as marathons or triathlons, many exercise physiologists believe that fatigue is the result of depleting energy stores within the body, and the physiological causes of fatigue are discussed in detail in this chapter. Alternatively, the *psychobiological model of endurance performance* suggests that fatigue is caused by the conscious decision to terminate a given activity, rather than by physiological limits. According to this model, the ultimate determinant of endurance performance in highly motivated athletes is the conscious perception of how hard, heavy, and strenuous the effort is. Based on this reasoning, strategies to reduce the *perception* of effort may delay the onset of fatigue in endurance athletes.

One strategy that may reduce fatigue according to the psychobiological model of endurance performance is *self-talk*, or self-addressed verbalizations, either aloud or silently. Self-talk can be both instructional and motivational for athletes, and it has been suggested to improve performance on effort-based tasks by motivating athletes to push themselves further even when the perceptual drive to terminate exercise is high. A 2014 study of 24 recreationally active men and women investigated the effect of self-talk on endurance performance during high-intensity cycling exercise.³ After baseline data were recorded, research subjects were divided into two groups: One group received 2 weeks of coaching and practice in the use of self-talk while the other group (control group) did not. After the 2 weeks, all of the participants returned to the laboratory for retesting, and their performance

results were compared to those from the beginning of the study. The group of subjects who received the self-talk coaching had a lower rating of perceived exertion (RPE) and a longer time to exhaustion (by almost 2 min) compared to their first visit (see figure); however, the results for the control group did not change. Motivational self-talk can reduce the perceived effort and increase endurance performance during aerobic activity. Training with psychobiological interventions that reduce the perception of effort may improve endurance performance in endurance athletes by delaying fatigue.



Changes in time to exhaustion after 2 weeks in the control group (dotted line) and the group of subjects who received 2 weeks of self-talk training (solid line). The self-talk group showed improvements in endurance after the self-talk training, while the control group did not change across the 2 weeks.

They found that when the electrical stimulation was applied after fatigue, the muscle was again able to maintain 20% MVC, indicating that, given the proper neural stimulation, the muscle itself maintains the ability to contract and generate force. When they recorded the muscle's electrical activity during the MVC after fatigue, they found that muscle activation by the nervous system *increased* greatly, suggesting that the reduced force of the maximal contraction was due to impairments of the contractile elements. Overall, this study suggests that the initial fatigue experienced after a bout of submaximal exercise is likely due to a reduction in central neural factors, whereas the impairments in maximal contraction are due to peripheral factors related to changes in excitation–contraction coupling.

Undoubtedly, there is some central nervous system (CNS) involvement in most types of fatigue. When a subject's muscles

appear to be nearly exhausted, verbal encouragement, shouting, music, or even direct electrical stimulation of the muscle can increase the strength of muscle contraction. The precise mechanisms underlying the CNS role in causing, sensing, and even overriding fatigue are not fully understood. Unless they are highly motivated, most individuals terminate exercise before their muscles are physiologically exhausted. To achieve peak performance, athletes train to learn proper pacing and tolerance for fatigue.



Other Contributors to Fatigue

As one can appreciate from the previous sections, the underlying causes of fatigue are many and varied and depend largely on the

intensity and duration of the exercise being performed. Recent research has also uncovered roles for impaired mitochondrial function and reactive oxygen species in some types of fatigue. A group of French investigators reported that impaired mitochondrial function led to both a slower rate of PCr recovery and a reduction in oxidative ATP production after dynamic submaximal exercise.¹⁰ However, the extent to which those effects can be attributed to cellular acidosis or muscle damage is unclear.

Michael Reid provided a compelling case that reactive oxygen species (ROS) accumulation in working muscle contributes to the loss of function that occurs in muscle fatigue.²⁵ These molecules, including hydrogen peroxide, superoxide, and hydroxyl radicals, increase during strenuous muscle contractions. They are present in myofibrillar cytoplasm and organelles, in interstitial fluid, and within the vascular space. Two lines of evidence point to a role for these chemically reactive molecules: (1) Directly exposing muscle cells to ROS evokes many of the same changes that occur with fatigue during exercise and (2) pretreating the muscle with antioxidants delays fatigue.

In Review

- Depending on the circumstances, fatigue may result from depletion of PCr or glycogen; both situations impair ATP production. Glycogen depletion may occur in select fiber types or specific muscle groups depending on the exercise.
- Increased metabolic by-products like phosphate ions and heat may contribute to fatigue.
- Lactic acid often has been blamed for fatigue, but it is generally not directly related to fatigue during prolonged endurance exercise, and may serve as a fuel source (see [chapter 2](#)).
- In short-duration exercise, like sprinting, the H^+ generated by dissociation of lactic acid may contribute to fatigue. The accumulation of H^+ decreases muscle pH, which impairs the cellular processes that produce energy and muscle contraction.
- Failure of neural transmission may be a cause of some types of fatigue. Many mechanisms can lead to such failure, and further research is needed.

- The CNS plays a role in most types of fatigue, perhaps limiting exercise performance as a protective mechanism. Perceived fatigue usually precedes physiological fatigue, and athletes who feel exhausted can often be encouraged to continue by various cues that stimulate the CNS, such as music or self-talk.
-

Critical Power: The Link Between Energy Expenditure and Fatigue

An athlete who can sustain a high level of exercise intensity for a prolonged period without fatiguing will be successful. Exercise physiologists have a name for the link between optimal performance and fatigue: **critical power**. The critical power defines the tolerable duration of high-intensity exercise. If we graph the relation between power output (or exercise intensity, or speed) and the maximal time that intensity can be maintained, the line is curvilinear, as depicted in [figure 5.13](#). At very high power outputs, exercise can be performed only for short durations. But as intensity is progressively decreased, exercise can be performed for longer and longer durations. At some point, this relation levels off and reaches an asymptote, defining the critical power for that activity—the maximal intensity that can be sustained without fatigue limiting performance.

Critical power represents the highest metabolic rate that is maintained entirely by oxidative metabolism. In that regard, it is related to the lactate threshold (discussed earlier in this chapter), but occurs at slightly higher intensities. Not surprisingly, critical power is increased with endurance or high-intensity interval training and decreased with aging and in chronic disease states. Hypoxia, such as that encountered at altitude (discussed in [chapter 13](#)), also reduces critical power, while breathing elevated oxygen concentrations elevates it.

Critical power is a useful measure in sport and exercise physiology because it correlates well with performance in running, rowing, swimming, and even team sport activities lasting from a few min to 2 h.²⁸ However, while exercise at or below the critical power should theoretically be able to be continued indefinitely, in reality exercise at the critical power cannot be sustained beyond 30 min or so. With much attention being paid to breaking the 2 h barrier for the

marathon, the critical power concept would dictate that a runner has to sustain a critical speed of only 21.1 km/h (13.1 mph, or slightly under 4.6 min miles); however, sustaining that heavy-intensity pace for 2 h has proven virtually impossible.²⁸

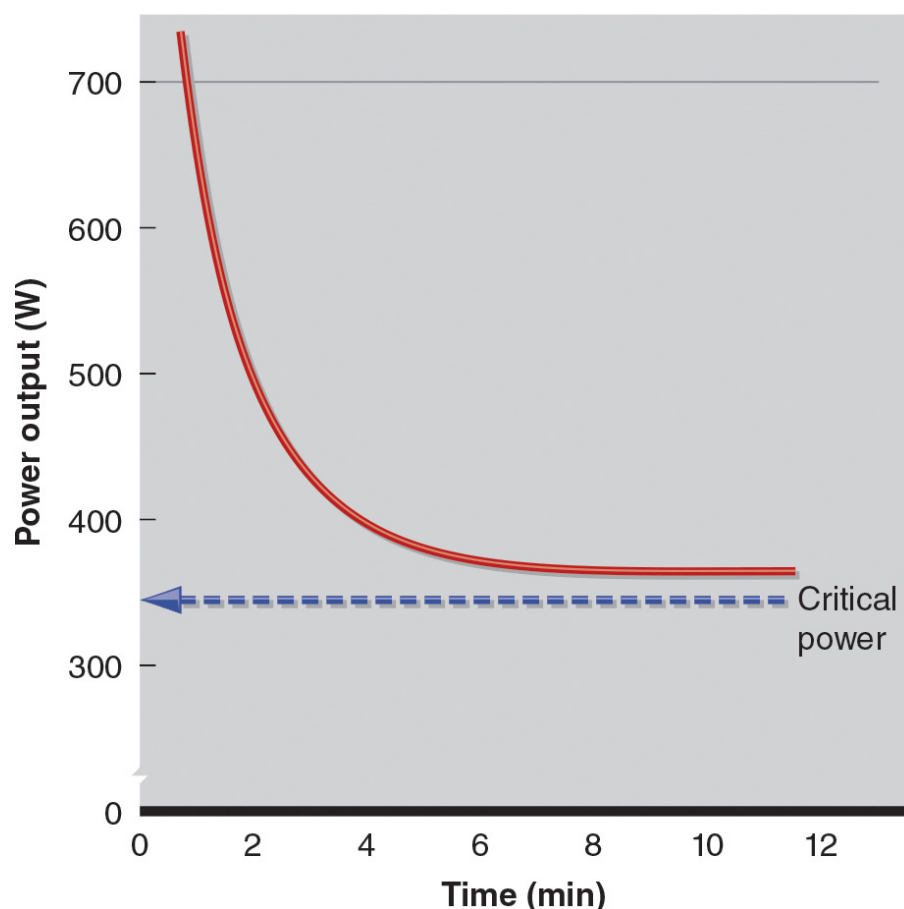


FIGURE 5.13 The relation between power output (in watts [W]) and the time that power output can be maintained. The critical power is defined as the asymptote in the relation, i.e., the maximal power output that can be sustained without fatigue limiting the duration of performance.



Muscle Soreness and Muscle Cramps

Muscle soreness generally results from exercise that is exhaustive or of very high intensity. This is particularly true when people perform a

specific exercise for the first time. While muscle soreness can be felt at any time, there is generally a period of mild muscle soreness that can be felt during and immediately after exercise and then a more intense soreness felt a day or two later.

Acute Muscle Soreness

Pain felt during and immediately after exercise is classified as a muscle strain and is perceived as muscle stiffness, aching, or tenderness. It can result from accumulation of the end products of exercise, such as H^+ , and from tissue edema that is caused by fluid shifting from the blood plasma into the tissues. Edema is the cause of the acute muscle swelling that people feel after heavy endurance or strength training. The pain and soreness usually disappear within several hours after the exercise. Thus, this soreness is often referred to as **acute muscle soreness**.

Delayed-Onset Muscle Soreness

The precise causes of muscle soreness felt a day or two after a heavy bout of exercise are not totally understood. Because this pain does not occur immediately, it is referred to as **delayed-onset muscle soreness (DOMS)**. Delayed-onset muscle soreness can vary from slight muscle stiffness to severe, debilitating pain that restricts movement. In the following sections, we discuss some theories that attempt to explain this form of muscle soreness.

RESEARCH PERSPECTIVE 5.3

Are Muscle Fatigue and Exercise Inefficiency the Same Thing?

During whole-body exercise, fatigue and decreased efficiency (the ratio of mechanical energy output, or external work performed, to metabolic energy production) are major causes of exercise intolerance and resulting early termination of an acute exercise bout. Exercise physiologists agree that these two concepts, fatigue and decreased efficiency, contribute to exercise intolerance, but are they linked?

Decreased efficiency typically precedes exercise termination during high-intensity exercise. There is an increased oxygen cost of work during constant power output and incremental exercise above the lactate threshold. This is most clearly seen with $\dot{V}O_2$ drift during steady-state exercise, where $\dot{V}O_2$

slowly increases despite a constant power output; this increased oxygen cost for the same amount of work is evidence of a decreased efficiency of muscle contraction. Similarly, during incremental exercise, $\dot{V}O_2$ increases in excess of predicted need for power outputs above the lactate threshold. In fact, during incremental exercise, efficiency is reduced by approximately 20% for power outputs above the lactate threshold. As a consequence of this inefficiency, the exerciser reaches his or her peak $\dot{V}O_2$ at a lower power output, resulting in muscle fatigue and ultimately failure to continue. Overall, the decline in skeletal muscle efficiency during high-intensity exercise above the lactate threshold dictates a greater oxygen demand to produce the same mechanical power, termed inefficiency. Because the energy produced by the muscle is limited, the rate at which this inefficiency develops is a major determinant of fatigue and task failure.

But what is causing muscle inefficiency at high power outputs? The fact that muscle fatigue during whole-body exercise occurs only at intensities above the lactate threshold (where ATP production relies on contributions from substrate-level phosphorylation) may provide clues, and the factors affecting the ratio between ATP resynthesis and oxygen consumption by the muscle mitochondria may provide the answer. A 2014 study using combined magnetic resonance spectroscopy and pulmonary gas exchange in humans found that the tight relation between ATP production and $\dot{V}O_2$ that is observed at moderate-intensity exercise was lost at exercise intensities above the lactate threshold.⁵ This finding suggests that muscle inefficiency is due to impairments in ATP production and turnover.

As discussed in this chapter, many intracellular mechanisms, including changes in oxygen and substrate availability, impaired function of the ATPases, decreased pH, increased temperature, altered Na^+/K^+ pump function, and changes in motor unit recruitment patterns have all been studied and shown to contribute to reduced muscle efficiency and muscle fatigue. Impairments in ATP turnover and production challenge the cellular homeostasis of the muscle fiber. In this scenario, muscle fatigue may be a protective mechanism that prevents muscle fiber damage. Changes in muscle cellular processes during high-intensity aerobic exercise provide a link between this inefficiency and muscle fatigue that ultimately leads to failure.¹³ However, no one has shown a clear cause–effect relation between muscle fatigue and decreased efficiency. Future research is necessary to tell if fatigue and inefficiency are in fact the same thing.

Almost all current theories acknowledge that eccentric muscle action is the primary initiator of DOMS. This has been clearly demonstrated in a number of studies examining the relationship of muscle soreness to eccentric, concentric, and static actions.

Individuals who train solely with eccentric actions experience extreme muscle soreness, whereas those who train using only static and concentric actions experience little soreness. This idea has been further explored in studies in which subjects ran on a treadmill for 45 min on two separate days, one day on a level grade and the other day on a 10% downhill grade.^{26,27} No muscle soreness was associated with the level running. But the downhill running, which required extensive eccentric action, resulted in considerable soreness within 24 to 48 h, even though blood lactate concentrations, previously thought to cause muscle soreness, were much higher with level running.

In the following sections we examine some of the proposed explanations for exercise-induced DOMS. In general, the pathway for developing DOMS begins with structural damage to muscle fibers (microtrauma) and to the surrounding connective tissues. This damage is followed by an inflammatory process that leads to edema as fluid and electrolytes shift into the area. To make matters worse, muscle spasms can occur, prolonging the condition and making the soreness worse.

Structural Damage

The presence of increased concentrations of several specific muscle enzymes in blood after intense exercise suggests that some structural damage may occur in the muscle membranes. These enzyme concentrations in the blood increase from 2 to 10 times following bouts of heavy training. Recent studies support the idea that these changes might indicate various degrees of muscle tissue breakdown. Examination of tissue from the leg muscles of marathon runners has revealed remarkable damage to the muscle fibers after both training and marathon competition. The onset and timing of these muscle changes parallel the degree of muscle soreness experienced by the runners.

Figure 5.14 shows changes in the contractile filaments and Z-disks before and after a marathon race. Recall that Z-disks are the points of contact for the contractile proteins. They provide structural support for the transmission of force when the muscle fibers are activated to shorten. Figure 5.14b, after the marathon, shows

moderate Z-disk streaming and major disruption of the thick and thin filaments in a parallel group of sarcomeres as a result of the force of eccentric actions or stretching of the tightened muscle fibers.

Although the effects of muscle damage on performance are not fully understood, it is generally agreed that this damage is responsible in part for the localized muscle pain, tenderness, and swelling associated with DOMS. However, blood enzyme concentrations might increase and muscle fibers might be damaged frequently during daily exercise that produces no muscle soreness. Also, remember that muscle damage appears to be a precipitating factor for muscle hypertrophy.

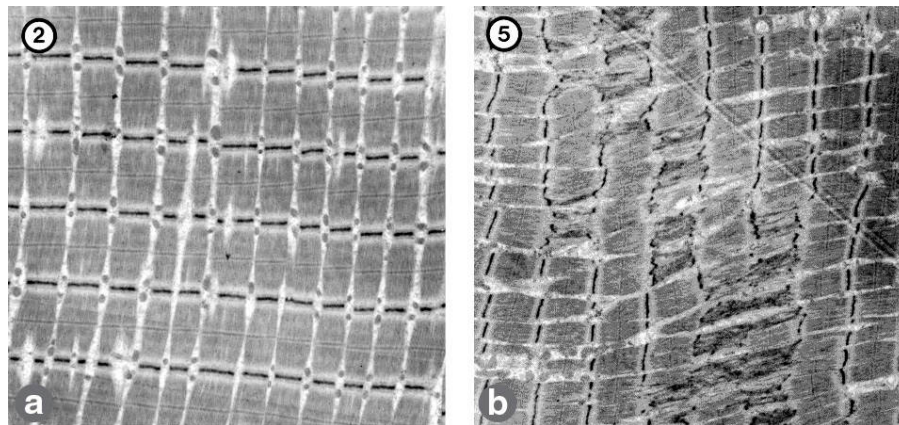


FIGURE 5.14 (a) An electron micrograph showing the normal arrangement of the actin and myosin filaments and Z-disk configuration in the muscle of a runner before a marathon race. (b) A muscle sample taken immediately after a marathon race shows moderate Z-disk streaming and major disruption of the thick and thin filaments in a parallel group of sarcomeres, caused by the eccentric actions of running.

Reprinted by permission from S.M. Roth et al., "High-Volume, Heavy-Resistance Strength Training and Muscle Damage in Young and Older Women," *Journal of Applied Physiology* 88 (2000): 1112-1118. Image courtesy of Dr. Roth.

Inflammatory Reaction

White blood cells serve as a defense against foreign materials that enter the body and against conditions that threaten the normal function of tissues. The white blood cell count tends to increase following activities that induce muscle soreness, leading some investigators to suggest that soreness results from inflammatory reactions in the muscle. But the link between these reactions and muscle soreness has been difficult to establish.

In early studies, researchers attempted to use drugs to block the inflammatory reaction, but these efforts were unsuccessful in reducing either the amount of muscle soreness or the degree of inflammation. These early results did not support a link between simple inflammatory mediators and DOMS. However, more recent studies have begun to establish a link between muscle soreness and inflammation. It is now recognized that substances released from injured muscle can act as attractants, initiating the inflammatory process. Mononucleated cells in muscle are activated by the injury, providing the chemical signal to circulating inflammatory cells. Neutrophils (a type of white blood cell) invade the injury site and release cytokines (immunoregulatory substances), which then attract and activate additional inflammatory cells. Neutrophils possibly also release oxygen free radicals that can damage cell membranes. The invasion of these inflammatory cells is also associated with the incidence of pain, thought to be caused by a release of substances from the inflammatory cells stimulating the pain-sensitive nerve endings. Macrophages (another type of cell of the immune system) then invade the damaged muscle fibers, removing debris through a process known as phagocytosis. Last, a second phase of macrophage invasion occurs, which is associated with muscle regeneration.

Sequence of Events in DOMS

The general consensus among researchers is that a single theory or hypothesis cannot explain the mechanism causing DOMS. Instead researchers have proposed a sequence of events that may explain the DOMS phenomenon, including the following:

1. High tension in the contractile-elastic system of muscle results in structural damage to the muscle and its cell membrane. This is also accompanied by excessive strain of the connective tissue.
2. The cell membrane damage disturbs calcium homeostasis in the injured fiber, inhibiting cellular respiration. The resulting high calcium concentrations activate enzymes that degrade the Z-lines.

3. Within a few hours there is a significant elevation in circulating neutrophils that participate in the inflammatory response.
4. The products of macrophage activity and intracellular contents (such as histamine, kinins, and K^+) accumulate outside the cells. These substances then stimulate the free nerve endings in the muscle. This process appears to be accentuated in eccentric exercise, in which large forces are distributed over relatively small cross-sectional areas of the muscle.
5. Fluid and electrolytes shift into the area, creating edema, which causes tissue swelling and activates pain receptors. Muscle spasms may also be present.

DOMS and Performance

With DOMS comes a reduction in the force-generating capacity of the affected muscles. Whether the DOMS is the result of injury to the muscle or edema, the affected muscles are not able to exert as much force when the person is asked to apply maximal force, as in the performance of a 1-repetition maximum strength test. Maximal force-generating capacity gradually returns over days or weeks. The loss of strength is the result of

1. the physical disruption of the muscle as illustrated in [figure 5.14](#),
2. failure within the excitation–contraction coupling process, and
3. loss of contractile protein.

Failure in excitation–contraction coupling appears to be the most important, particularly during the first 5 days. This is illustrated in [figure 5.15](#).

Muscle glycogen resynthesis also is impaired when a muscle is damaged. Resynthesis is generally normal for the first 6 to 12 h after exercise, but it slows or stops completely as the muscle undergoes repair, thus limiting the fuel storage capacity of the injured muscle. [Figure 5.16](#) illustrates the time sequence of the various markers of muscle damage associated with intense eccentric exercise of the elbow flexor muscles as compared to concentric exercise. As shown in the figure, changes in function (MVC and range of motion), muscle swelling (circumference), soreness, and molecular indicators of

damage (creatine kinase activity and myoglobin concentration) persist for several days.

Minimizing DOMS

Reducing the negative effects of DOMS is important for maximizing training gains. The eccentric component of muscle action could be minimized during early training, but this is not possible for athletes in most sports. An alternative approach is to start training at a very low intensity and progress slowly through the first few weeks. Yet another approach is to initiate the training program with a high-intensity, exhaustive training bout. Muscle soreness would be great for the first few days, but evidence suggests that subsequent training bouts would cause considerably less muscle soreness. Because the factors associated with DOMS are also potentially important in stimulating muscle hypertrophy, DOMS is most likely necessary to maximize the training response.

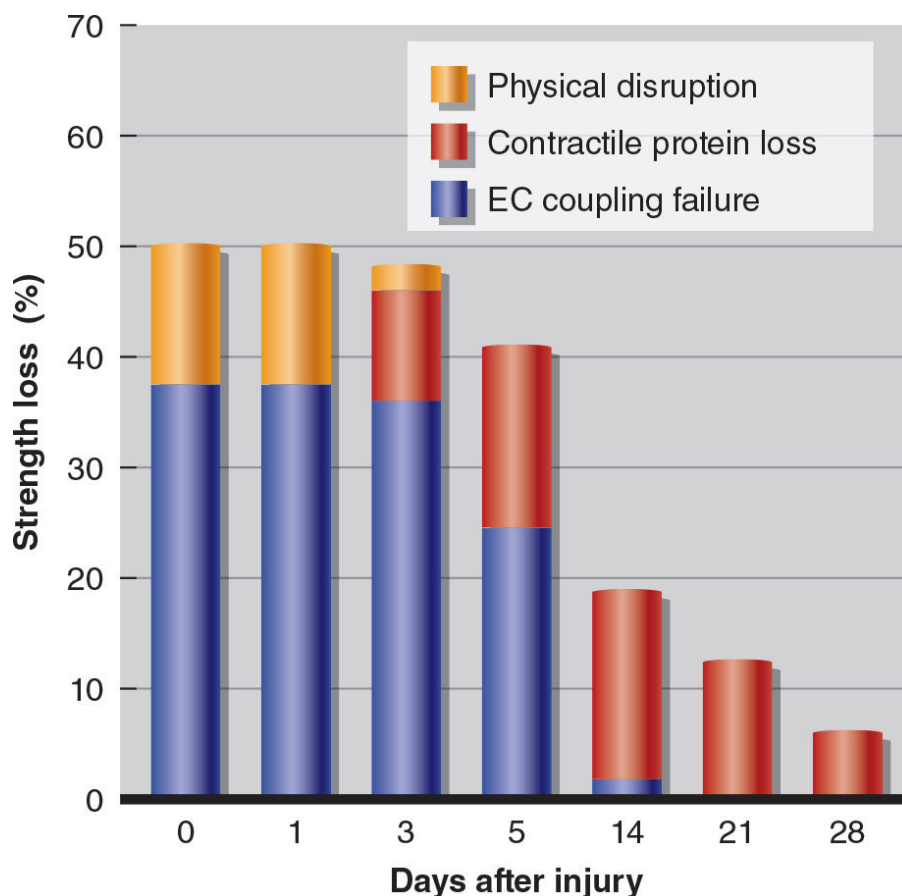


FIGURE 5.15 Estimated contributions of excitation–contraction (EC) coupling failure, decreased contractile protein content, and physical disruption to the decrease in strength following muscle injury. Reprinted by permission from G. Warren et al., “Excitation-Contraction Uncoupling: Major Role in Contraction-Induced Muscle Injury,” *Exercise and Sport Sciences Reviews* 29, no. 2 (2001): 82-87

Statins and Skeletal Muscle Soreness

3-Hydroxy-3-methylglutaryl (HMG)-CoA reductase inhibitors, or statins, are the most commonly prescribed cardiovascular drugs in the world. Statins are extremely effective at reducing serum cholesterol concentration and reducing the risk of future cardiovascular events. The most common side effect associated with taking statins is muscle pain, which is reported to occur in up to 25% of patients.²³ Muscle pain from statins can range from mild soreness including cramps and weakness to a life-threatening condition associated with a severe breakdown of muscle tissue called rhabdomyolysis. While the precise mechanism for how statins may contribute to muscle soreness and damage is unclear, it has been linked to excessive production of reactive oxygen molecules by the mitochondria and changes in the way muscle cells get rid of damaged proteins.

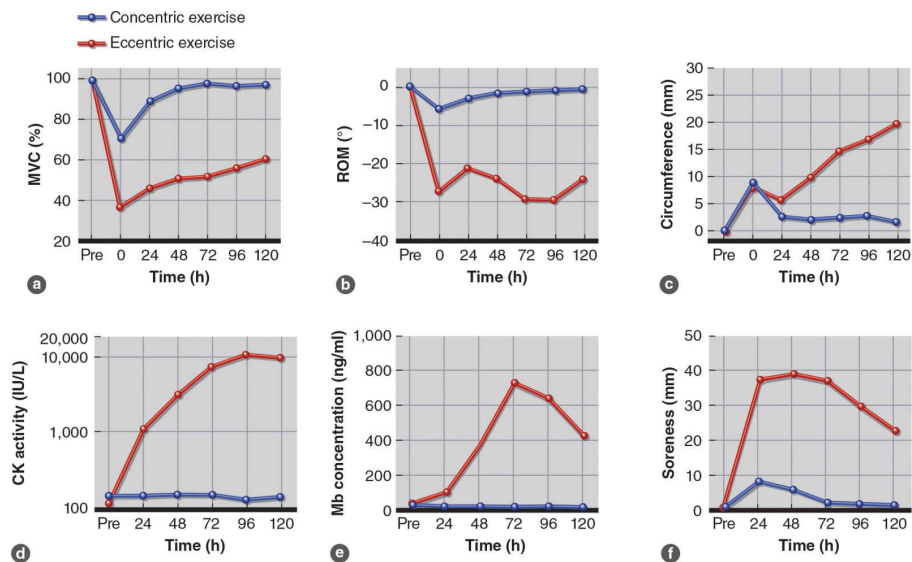


FIGURE 5.16 The responses of various physiological markers of muscle damage after eccentric and concentric exercise by elbow flexors. The changes persist for several days and include (a) MVC and (b) range of motion (ROM), both indicators of muscle function; (c) muscle swelling (circumference); (d) creatine kinase (CK) and (e) plasma myoglobin (Mb) concentration, both molecular indicators of damage; and (f) soreness.

Reprinted by permission from K. Nosaka, Muscle Soreness and Damage and the Repeated-Bout Effect, in *Skeletal Muscle Damage and Repair*, edited by Peter Tiidus (Champaign, IL: Human Kinetics, 2008). Data from A.P. Lavender and K. Nosaka, "Changes in Steadiness of Isometric Force Following Eccentric and Concentric Exercise," *European Journal of Applied Physiology* 96 (2006): 235-240.

Statin use increases creatine kinase concentration after eccentric exercise, a clinical marker of muscle protein breakdown. However, patients who take statins may have muscle pain without an increase in creatine kinase, suggesting that other mechanisms might be causing pain.²² While trained individuals may be able to tolerate pain during vigorous exercise, in some people, statin-associated muscle pain may limit even leisure-type physical activity.⁴ Additionally, recent exercise training studies in older people indicate that statin users do not fully adapt to the training stimulus.¹⁷ Because exercise is a cornerstone therapy for treating and preventing cardiovascular disease, much more research needs to be done to more fully understand the effects of statins on skeletal muscle physiology and how to optimize the beneficial effects of both therapies.

In Review

- Acute muscle soreness occurs during or immediately after an exercise bout.
- Delayed-onset muscle soreness usually peaks a day or two after the exercise bout. Eccentric muscle action seems to be the primary initiator of this type of soreness.
- Proposed causes of DOMS include structural damage to muscle cells and inflammatory reactions within the muscles. The proposed sequence of events includes structural damage, impaired calcium homeostasis, inflammatory response, increased macrophage activity, and edema.
- Reduced muscle strength with DOMS is likely the result of physical disruption of the muscle, failure of the excitation–contraction process, and loss of contractile protein.
- Muscle soreness can be minimized through the use of lower intensity and fewer eccentric contractions early in training. However, muscle soreness may ultimately be an important part of maximizing the resistance training response.

RESEARCH PERSPECTIVE 5.4

Delayed-Onset Muscle Soreness May Be Different in Men and Women

Creatine kinase is the enzyme that catalyzes the exchange of high-energy phosphate bonds between phosphocreatine and ADP to supply ATP to the working muscle during exercise. When creatine kinase appears in the blood, it can indicate metabolic and mechanical disturbances in the muscle cell. Following eccentric exercise in men, creatine kinase activity measured in the blood correlates with muscle soreness and decrements in maximal isometric strength. Men have higher blood creatine kinase activity compared to women, which may be due to the actions of circulating estrogens. The creatine kinase response to exercise may be lower when women are tested during periods of higher circulating estrogen concentration (e.g., the late follicular phase preceding ovulation) compared to periods of low circulating estrogen. Some studies of sex differences in reported muscle soreness suggest that women have a reduced perception of soreness following eccentric exercise compared to men, but other studies have found no differences. Interestingly, the creatine kinase response to exercise is correlated with delayed-onset muscle soreness (DOMS) in men but not in women. Altogether, there is still debate about whether sex differences in the creatine kinase response to exercise, and in DOMS, exist in humans.

A recent study conducted in South Africa set out to determine whether the serum creatine kinase response and the perception of DOMS following a bout of downhill running were influenced by sex and if the magnitudes of those responses depended on the circulating estrogen concentrations in the women.¹⁹ In that study, 21 sedentary subjects (6 men and 15 women) performed 20 min of downhill running on a treadmill in the laboratory. Blood samples were collected before exercise, immediately after exercise, and then 24, 48, and 72 h after exercise. Blood samples were analyzed for creatine kinase activity and for estrogen and progesterone concentrations in the women. The researchers also assessed DOMS at the same time points by having the subjects rate their perception of soreness when standing from a seated position on a visual scale from “no pain” to “the worst pain ever experienced.”

The 24 h peak creatine kinase response to this bout of downhill running was the same between men and women; however, circulating creatine kinase activity was restored to preexercise baseline faster in women (by 48 h postexercise) than in men (72 h after exercise). Neither estrogen nor progesterone influenced the creatine kinase response in the women. Interestingly, both men and women still reported muscle soreness at 72 h after the eccentric exercise bout despite the recovery of creatine kinase in women 24 h earlier. While feelings of muscle soreness were prolonged in women who participated during the follicular phase of their menstrual cycle, the researchers were not able to determine if the associated hormones (estrogen or progesterone) were responsible for those findings.

Overall, the study concluded that both creatine kinase and DOMS responses to downhill running are affected by sex. Creatine kinase recovers more quickly in women, regardless of circulating reproductive hormone concentrations, but the recovery of muscle soreness is only correlated with creatine kinase concentrations in men. While the DOMS response in women may be affected by menstrual phase, a direct link to circulating hormones has not been demonstrated.

Exercise-Induced Muscle Cramps

Skeletal muscle cramps are a frustrating problem in sport and physical activity and commonly occur even in highly fit athletes. Skeletal muscle cramps can come during the height of competition, immediately after competition, or at night during deep sleep. Muscle cramps are equally frustrating to scientists, because there are multiple and unknown triggers and causes of muscle cramping, and little is known about the best treatment and prevention strategies. Nocturnal muscle cramps, especially in the calf muscle, have been experienced by 60% of adults. This type of cramp is probably caused by muscle fatigue and nerve dysfunction and may or may not be associated with exercise. Electrolyte imbalances and hydration do not seem to play an important role.

Exercise-associated muscle cramps (EAMCs), on the other hand, are defined as painful, spasmodic, involuntary contractions of skeletal muscles that occur during or immediately after exercise. Two distinct theories have been proposed to explain the causes of EAMCs, termed the *neuromuscular control theory* and the *electrolyte depletion theory*.²

The neuromuscular control theory proposes that EAMCs occur when some aspect of control between the motor neuron and the muscle itself becomes altered. As muscle fatigue develops, excitation of the muscle spindle and inhibition of the Golgi tendon organ occur, resulting in abnormal α -motor neuron activity and reduced inhibitory feedback. This abnormal firing of motor neurons initially presents as muscle twitches or prefasciculations. If muscle contraction continues, an EAMC occurs.

Risk factors associated with this type of cramping are age, cramping history, and excessive exercise intensity and duration.

Several lines of evidence support this theory:

1. This type of cramping is generally localized to the overworked muscle.
2. Lack of conditioning, improper training, and depletion of muscle energy stores, which are all associated with muscle fatigue, can lead to the development of EAMCs.
3. Muscle cramping can be induced in the laboratory by electrical stimulation or voluntary muscle contraction (with no changes in electrolytes), which suggests that the mechanism is neuromuscular in origin.
4. Often the most effective treatment for relieving cramps is stretching of the muscle. Stretching increases tension in the muscle and in the Golgi tendon organ that inhibits the α -motor neuron.
5. Changing excitatory properties of the motor neuron—for example, by ingesting transient receptor potential (TRP) channel agonists—has been efficacious in attenuating both electrically induced and voluntarily induced EAMCs.⁷

The second theory, electrolyte depletion, better describes a different type of exercise-associated muscle cramp, often called *heat cramps*. This type of muscle cramp typically occurs in athletes who have been sweating extensively and have significant electrolyte disturbances, mainly of sodium and chloride. These cramps involve large muscle groups, and their occurrence is sometimes described as “locking up.” This type of exertional heat cramp usually evolves from small localized visible muscle fasciculations to severe and debilitating muscle spasms. The cramps often begin in the legs but can become widespread.

Because a significant amount of sodium can be lost only along with a large loss of fluid, this theory is typically coupled with dehydration. Progressive dehydration and electrolyte depletion cause fluid to shift from the interstitial compartment to the intravascular compartment. This contracts the extracellular fluid compartment, increasing surrounding neurotransmitter concentrations and causing selected motor nerve terminals to become hyperexcitable, leading to spontaneous discharge, initiation of action potentials in the muscles, and ultimately EAMCs.

Proponents of this theory have espoused the following:

1. Anecdotal evidence has existed for centuries that laborers working in hot and humid conditions suffered from cramping.
2. In those laborers, ingesting small volumes of salt water prevented or alleviated the cramps.
3. Increases in sweat sodium concentration (“salty sweating”) is evident in those athletes, specifically tennis and American football players, who are most prone to cramping.⁹

VIDEO 5.1 Presents Mike Bergeron on the two types of muscle cramping and the best ways to prevent muscle cramps.



Treatment of heat cramps involves the prompt ingestion of a high-salt solution (3 g in 500 ml of a sodium-containing beverage every 5 to 10 min) or intravenous fluid and sodium loading. In addition, massage and ice application may help to calm the affected muscles and relieve pain. Electrolyte-containing fluids should be continued if dehydration and electrolyte loss are suspected.

To prevent EAMCs, the athlete should

- be well conditioned, to reduce the likelihood of muscle fatigue;
- regularly stretch the muscle groups prone to EAMCs;
- maintain fluid and electrolyte balance and carbohydrate stores; and
- reduce exercise intensity and duration if necessary.

In Review

- Muscle fatigue-associated cramps are related to sustained α -motor neuron activity, with increased muscle spindle activity and decreased Golgi tendon organ activity.
 - Exercise-associated muscle cramps may be caused by altered neuromuscular control, fluid or electrolyte imbalances, or both.
 - Heat-associated cramps, which typically occur in athletes who have been sweating excessively, involve a shift in fluid from the interstitial space to the intravascular space, resulting in a hyperexcitable neuromuscular junction.
 - Rest, passive stretching, holding the muscle in the stretched position, and fluid and electrolyte restoration can be effective in treating EAMCs. Proper conditioning, stretching, and nutrition are also possible prevention strategies.
-

IN CLOSING

In previous chapters, we discussed how muscles and the nervous system function together to produce movement. In this chapter we focused on energy expenditure during exercise and fatigue. We considered the energy needed by the body at rest and during movement. We explored how energy production and availability can limit performance and learned that metabolic needs vary considerably. We discussed the many potential factors involved in fatigue, including those resulting from decreased energy delivery and accumulation of metabolic by-products and those associated with the peripheral and central nervous systems. We also introduced the concept of critical power as a link between energy expenditure and fatigue. We also examined delayed-onset muscle soreness and muscle cramps as additional limiting factors in exercise. In the next chapter, we turn our attention to the cardiovascular system and its control.

KEY TERMS

acute muscle soreness
basal metabolic rate (BMR)
calorie (cal)
calorimeter
cardiorespiratory endurance
central governor theory
critical power
delayed-onset muscle soreness (DOMS)
direct calorimetry
excess postexercise oxygen consumption (EPOC)
exercise-associated muscle cramps (EAMCs)
fatigue
Haldane transformation
indirect calorimetry
lactate threshold
maximal oxygen uptake ($\dot{V}O_{2\max}$)
oxygen deficit
peak oxygen uptake ($\dot{V}O_{2\text{peak}}$)
respiratory exchange ratio (RER)
resting metabolic rate (RMR)
 $\dot{V}O_2$ drift

STUDY QUESTIONS

1. Define direct calorimetry and indirect calorimetry, and describe how they are used to measure energy expenditure.
2. What is the respiratory exchange ratio (RER)? Explain why it is used to determine the relative contributions of carbohydrate and fat to energy expenditure.
3. What are basal metabolic rate and resting metabolic rate, and how do they differ?
4. What is maximal oxygen uptake? How is it measured? What is its relationship to sport performance?
5. Describe two possible markers of anaerobic capacity.
6. What is the lactate threshold? How is it measured? What is its relation to sport performance?
7. What is economy of effort? How is it measured? What is its relationship to sport performance?
8. What is the relationship between oxygen consumption and energy production?
9. Why do athletes with high $\dot{V}O_{2\max}$ values perform better in endurance events than those with lower values?
10. Why is oxygen consumption often expressed as milliliters of oxygen per kilogram of body weight per minute ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$)?
11. Describe the possible causes of fatigue during exercise bouts lasting 15 to 30 s and those lasting 2 to 4 h.
12. Discuss three mechanisms through which lactate can be used as an energy source.
13. Define critical power and explain its usefulness in sport physiology. What is its relation to sport performance?
14. What is the physiological basis for delayed-onset muscle soreness?
15. What two theories have been proposed to explain the physiological basis for exercise-associated muscle cramps? Provide support for each.

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.